Jongmin Lee

Position

Assistant Professor
Department of Computer Science and Engineering
School of Computer Science and Engineering, College of Software
Chung-Ang University (CAU)

Contact Information

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Education

Pohang University of Science and Technology (POSTECH), Pohang, Korea

Ph.D., Dep. of Computer Science and Engineering (CSE) **Sep 2018 – Aug 2024**

- Advisor: Prof. Minsu Cho
- Thesis: *Learning Equivariant Features for Visual Correspondence and Pose Estimation*

Graz University of Technology (TU Graz), Graz, Austria **Sep 2016 – Feb 2017**

Pohang University of Science and Technology (POSTECH), Pohang, Korea

B.S., Dep. of Industrial and Management Engineering (IME)
B.S., Dep. of Computer Science and Engineering (CSE) **Mar 2013 – Aug 2018**

- Double Major

Work Experience

Chung-Ang University (CAU), Heukseok, Seoul, Korea.
Assistant Professor **Sep 2025 – Now**

- Teaching undergraduate/graduate courses in Computer Science & Artificial Intelligence.
- Leading the Spatial Intelligence Lab, with a focus on realizing real-world AI
- Research: 3D computer vision, machine learning, multimodal reasoning, and robotics.
- Lab hompage: https://bluedream1121.github.io/spatial-intelligence-lab/

LG AI Research., Magok, Seoul, Korea.
Research Scientist **Oct 2024 – Aug 2025**

- Team: Vision Lab. | Team: Exaone-VL TF
- Working on post-training of M-LLMs (SFT and RL-based methods DPO & GRPO)
- Exploring vision-language foundation models across perception and reasoning
- Results: Achieving SOTA results on AGI benchmarks (MMMU, AI2D, ChartQA, DocQA, etc.) at Jul. 2025, and contributing to deploy the in-house Exaone-4.0-VL models

Meta Reality Labs., Burlingame, California, U.S.A.
Ph.D Research Scientist Intern **Jul 2023 – Oct 2023**

- Team: XR Eyes | Manager: Survi Kyal, Mentor: Fengting Yang
- Research project: advancving eye tracking production model for Meta Quest Pro.
- Method: a self-supervised pre-training for invariant feature learning from large-scale multi-view video data.
- Results: Improved KPI metrics (P95, PCD < 4.5) by 9% in the Arcata project.

Vuno Inc., Gangnam, Seoul, Korea
Front-end Developer **Jun 2017 – Aug 2017**

- Developed the client interface for an AI software for diagnosis from a chest X-ray.

International Publications

Jongmin Lee, "Pair-Relative Partial Equivariance for Rotation-Robust Keypoint Descriptor Matching," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* **(WACV)**, 2027.

Yunchan Jeon*, SooGon Kim*, Seungwook Kim, CheolWon Lee, **Jongmin Lee**, "Beyond Family Labels: A Taxonomic Ornstein–Uhlenbeck Prior for Avian 3D Shape Recovery," in *the Fortieth Annual Conference on Neural Information Processing Systems* **(NeurIPS)**, 2026. **(Oral Presentation, 0.36% acceptance rate)**

Seungwook Kim, **Jongmin Lee**, "ProxySearch: Decoupled Inference-Time Scaling for Diffusion Models via Asymmetric Noise-Rank Transfer," in *the Fortieth Annual Conference on Neural Information Processing Systems* **(NeurIPS)**, 2026.

Anthony Cioppa, ..., Youngseon Kim, **Jongmin Lee**, et al., "SoccerNet 2026 Challenges Results," in *Proceedings of the Asian Conference on Computer Vision* **(ACCV)**, 2026.

Kwonyoung Ryu, In-Jae Lee, Jonghyun Jin, Hyunjee Lee, **Jongmin Lee**, Jaesik Park, "Extending a Large View Synthesis Model for Multi-view Panoptic Segmentation," in *Proceedings of the European Conference on Computer Vision* **(ECCV)**, 2026.

Seonghak Lee, Junhee Cho, Jisoo Park, Min-Gyu Park, **Jongmin Lee**, Ju Hong Yoon, Junseok Kwon, "URHead: A Unified UV-Space Representation for Joint Mesh–3DGS Optimization in Head Avatars," in *Proceedings of the European Conference on Computer Vision* **(ECCV)**, 2026.

Junhee Cho, Seonghak Lee, **Jongmin Lee**†, Junseok Kwon†, "DiNGGA: Dirichlet and Noise-Guided Gaussian Avatar for Fine-grained Facial Detail," *IEEE Access*, 2026. († co-corresponding authors)

Young Seon Kim, **Jongmin Lee**, "SetPieceRAG: Domain-Specific RAG for Knowledge-Intensive Soccer VQA with Large Language Models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* **(CVPRW)**, *Workshop on Computer Vision in Sports (CVSports)*, 2026. **(Spotlight)**

Jongmin Lee, Minsu Cho, "3D Equivariant Pose Regression via Direct Wigner-D Harmonics Prediction," in *the Thirty-eighth Annual Conference on Advances in Neural Information Processing Systems* **(NeurIPS)**, 2024.

Jongmin Lee, Byungjin Kim, Seungwook Kim, Minsu Cho, "Learning Rotation-Equivariant Features for Visual Correspondence," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* **(CVPR)**, 2023.

Jongmin Lee, Byungjin Kim, Minsu Cho, "Self-Supervised Equivariant Learning for Oriented Keypoint Detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* **(CVPR)**, 2022.

Jongmin Lee, Yoonwoo Jeong, Minsu Cho, "Self-supervised Learning of Image Scale and Orientation Estimation," in *Proceedings of the 32nd British Machine Vision Conference* **(BMVC)**, 2021.

Jongmin Lee, Yoonwoo Jeong, Seungwook Kim, Juhong Min, Minsu Cho, "Learning to Distill Convolutional Features Into Compact Local Descriptors," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* **(WACV)**, 2021.

Juhong Min, **Jongmin Lee**, Jean Ponce, Minsu Cho, "Learning to Compose Hypercolumns for Semantic Visual Correspondence," in *Proceedings of the European Conference on Computer Vision* **(ECCV)**, 2020.

Juhong Min, **Jongmin Lee**, Jean Ponce, Minsu Cho, "SPair-71k: A Large-scale Benchmark for Semantic Correspondence," arXiv preprint, 2019.

Juhong Min, **Jongmin Lee**, Jean Ponce, Minsu Cho, "Hyperpixel Flow: Semantic Correspondence with Multi-layer Neural Features," in *Proceedings of the IEEE/CVF International Conference on Computer Vision* **(ICCV)**, 2019.

Paul Hongsuck Seo, **Jongmin Lee**, Deunsol Jung, Bohyung Han, Minsu Cho, "Attentive Semantic Alignment with Offset-Aware Correlation Kernels," in *Proceedings of the European Conference on Computer Vision* **(ECCV)**, 2018.

Min Jung Lee, **Jongmin Lee**, Sanghyun Kim, Sunghyun Cho, Minsu Cho, "Base Frame Selection on Dynamically Exposed Burst for Multi-Image Restoration and Enhancement," in *Image Processing and Image Understanding* **(IPIU)** 2024.

Jongmin Lee, "Toward Spatial Intelligence: From Visual Geometry and 3D Vision," in *KIBME 2026 Summer Conference (Korean Institute of Broadcast and Media Engineers), Early-Career Researcher Session*, 2026.

Research Projects

POSTECH AI Research (PAIR) Feb 2024 - Aug 2024
- Video summarization with large language models.
- Modular reasoning models with LLaVA captioner, LLaMA caption aggregator.

Samsung Advanced Institute of Technology (SAIT) Nov 2022 - Jul 2023
- Burst image restoration with non-uniform exposure using base frame selection. (ISP)

Samsung Advanced Institute of Technology (SAIT) Nov 2021 - Oct 2022
- Burst image enhancement in large degradations by noise, blur and shift. (ISP Project)

Samsung Advanced Institute of Technology (SAIT) Nov 2020 - Oct 2021
- Motion-aware multiple image enhancement under extremely low-light conditions. (ISP)

Kakao Brain Nov 2021 - Jul 2023
- Efficient equivariant representation learning in deep neural networks.

Image Matching & Visual Localization Challenges Jul 2019 – Jun 2023
- Joined workshops at CVPR 2022/2023, ICCV 2019/2021, and ECCV 2020.
- Relevant links: Visual Localization Challenges | Image Matching Workshop

Lectures

Instructor

CS 39350 Computer Vision Applications @ CAU	Spring 2026
CS 38714 Creative Design @ CAU	Spring 2026
CS 59162 Deep Learning @ CAU	Fall 2025
CS 53744 Machine Learning Project @ CAU	Fall 2025
CS 40458 Automata and Formal Languages @ CAU	Fall 2025

Lecturer in online platform **Fast campus.**, Seoul, Korea Mar 2022 – Aug 2022
- Lectured on computer vision from classical to state-of-the-art deep learning approaches.
- Curriculum Sheet: Google Spreadsheet

Teaching assistant

Artificial Intelligence & Data Science (CSED537) @ POSTECH	Spring semester 2023
Introduction to Deep Learning (CSED490W) @ POSTECH	Fall semester 2019
POSCO AI expert training course	July 2019 – Aug 2019
Automata & Formal Languages (CSED341) @ POSTECH	Fall semester 2018

Professional Activities

Reviewer of international conferences

Computer Vision and Pattern Recognition (CVPR) 2022 – present
Neural Information Processing Systems (NeurIPS) 2023 – present
International Conference on Computer Vision (ICCV) 2023 – present
International Conference on Learning Representations (ICLR) 2024 – present
International Conference on Machine Learning (ICML) 2024 – present
European Conference on Computer Vision (ECCV) 2022 – present
Winter Conference on Applications of Computer Vision (WACV) 2021 – present
Asian Conference on Comoputer Vision (ACCV) 2024
International Conference on 3D Vision (3DV) 2022
British Machine Vision Conference (BMVC) 2021
Artificial Intelligence and Statistics Conference (AISTATS) 2025
International Conference on Machine Vision Applications (MVA) 2021, 2023
International Conference on Pattern Recognition (ICPR) 2020

Program committee of international conferences

Annual AAAI Conference on Artificial Intelligence (AAAI) 2025 – present

Reviewer of international journals

IEEE Transactions on Pattern Analysis and Machine Intelligence (2023 – present)	IF: 24.314
International Journal of Computer Vision (2023 – present)	IF: 19.500
IEEE Transactions on Image Processing (2022 – present)	IF: 11.041
Pattern Recognition (2022 – present)	IF: 8.518
The Visual Computer (2022)	IF: 2.835

Honours and Awards

Citations of Publications (2025): +500 citations in 9 papers in Google Scholar Profile

Open Source Contributions (2025): +300 stars in 10 projects in Github Profile

Qualcomm AI Innovation Fellowship Finalist (2023): Recognized as a finalist for this prestigious fellowship, reflecting exceptional innovation and potential in AI research. Awarded a grant of $1,000 by *Qualcomm Technologies Inc.*

BK21 Outstanding Paper Award (2022): Awarded for distinguished academic publication in the field of computer science engineering, Accompanied by a $500 grant.

Global Ph.D. Fellowship (2019–2024): Awarded by the *National Research Foundation of Korea (NRF)*, this significant fellowship ($120,000 grant) supports outstanding doctoral students demonstrating potential for becoming global leaders in their fields.

Excellent Research Award (2018): Honored by POSTECH Computer Science Engineering Department for superior research projects during the Undergraduate Research Program.

SK Hynix Scholarship (2015): Recognized for academic excellence and potential in technological innovation with a fellowship from the SK Hynix Fellowship Program.

Invited Talks

"Toward Spatial Intelligence", *Seoul National University (SNU) IPAI Seminar*, Seoul, Korea, November 2025. Host: Jaesik Park

"Toward Spatial Intelligence: Visual Geometry & 3D Vision", KAIST AI, Seoul, Korea, September 2025. Host: Seungryong Kim

"3D Equivariance: Learning Equivariant Features for Visual Correspondence and Pose Estimation", *Technion - Israel Institute of Technology*, Isreal, September 2024. Host: Or Litany

"Hyperpixel Flow: Semantic Correspondence with Multi-layer Neural Features", *ICCV 2019 Paper Day with Naver*, Hotel Andaz Gangnam, Seoul, Korea, October 2019.

"Where is semantic correspondence? - The general image matching problem in deep learning era", *Hyundai Motors AI Research Lab seminar*, Pohang, Korea, September 2019.

"Semantic Alignment - Find Semantic Dense Correspondence," *Naver corp.*, Pangyo, Korea, October 2018. Youtube link (Korean)

Language Skills

Korean(native), English(fluent)

Programming and Tools

Python, PyTorch, OpenCV, GitHub, HuggingFace, C, C++, MATLAB, SQL

Referees

Available on request.

Research Interests

My research interests mainly focus on developing novel models and algorithms to address practical challenges in deploying CV/ML systems to various real-world application domains. I am currently interested and engaged in the following topics:

- **Visual geometry & 3D vision:** visual correspondence, object/camera pose estimation, multi-view geometry, structure-from-motion, optical flow, 3D reconstruction, 3D symmetry, point cloud registration, robotics, embodied AI, and spatial reasoning.
- **Representation learning:** invariance and equivariance, geometric deep learning, equivariant learning, self-supervised learning, diffusion models for discriminative tasks, foundation models for image/video/3D.
- **Multi-modal large language models:** Vision-language applications for deep document understanding and video summarization, M-LLMs post-training (e.g., SFT, RLHF, and synthetic data), reinforcement fine-tuning or test-time adaptation for multi-modal applications, vision-language-action for intelligent robot and physical AI

The application domains of interest include, but are not limited to, **autonomous vision systems** (e.g., visual SLAM, autonomous driving, and industrial robotics), and **AR/VR technologies** (e.g., eye tracking, human-centric AI, wearable devices, and neural rendering).